

CSCE 763: Digital Image Processing

Homework #2

Instructor: Spring 2026
University of South Carolina
Solutions by: JC Vaught

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Problem 1: Median Operator Nonlinearity (10 pts)

The median ζ , of a set of numbers is such that half the values in the set are below ζ , and the other half are above it. For example, the median of the set of values $\{12, 3, 8, 20, 21, 75, 31\}$ is 20. Show that an operator that computes the median of a subimage area, S , is nonlinear.

Hints: an operator H is linear if

$$H[a_1f_1(x, y) + a_2f_2(x, y)] = a_1H[f_1(x, y)] + a_2H[f_2(x, y)]$$

Step 1: State the Linearity Condition

An operator H is said to be **linear** if and only if it satisfies the **superposition property** for all images f_1, f_2 and all scalar constants a_1, a_2 :

$$H[a_1f_1 + a_2f_2] = a_1H[f_1] + a_2H[f_2]$$

This property combines two fundamental requirements: *additivity* (the response to a sum equals the sum of responses) and *homogeneity* (scaling the input scales the output proportionally). For an operator to be linear, this equation must hold for *every possible* combination of inputs and scalars. Conversely, to prove an operator is nonlinear, we only need to find a *single* counterexample where this equality fails.

For simplicity, we will test the additivity condition by setting $a_1 = a_2 = 1$ and examining a small 1×3 subimage region S . If additivity fails, the full superposition property cannot hold, and the operator must be nonlinear.

Step 2: Define the Counterexample

We construct two simple images defined by their pixel values within the subimage region S . The key insight is to choose values where the median of each image individually differs significantly from the median of their sum:

$$f_1 = \{2, 3, 10\}, \quad f_2 = \{10, 1, 2\}$$

Notice that f_1 has its values spread across a range with the middle value being 3, while f_2 has its values arranged so that the middle value when sorted is 2. These particular values are chosen because when we add them together pixel-wise, the resulting set will have a very different median than the sum of the individual medians.

Step 3: Compute the Right-Hand Side

First, we compute the median of each image separately and then add the results. This corresponds to the right-hand side of the linearity equation, $H[f_1] + H[f_2]$.

For $f_1 = \{2, 3, 10\}$: sorting gives $\{2, 3, 10\}$, and the middle value is **3**.

For $f_2 = \{10, 1, 2\}$: sorting gives $\{1, 2, 10\}$, and the middle value is **2**.

Therefore:

$$H[f_1] + H[f_2] = \text{median}\{2, 3, 10\} + \text{median}\{10, 1, 2\} = 3 + 2 = \mathbf{5}$$

Step 4: Compute the Left-Hand Side

Now we compute the left-hand side of the linearity equation, $H[f_1 + f_2]$, by first adding the images pixel-wise and then taking the median of the result.

Adding corresponding pixel values:

$$f_1 + f_2 = \{2 + 10, 3 + 1, 10 + 2\} = \{12, 4, 12\}$$

Now we find the median of this combined image. Sorting gives $\{4, 12, 12\}$, and the middle value is:

$$H[f_1 + f_2] = \text{median}\{12, 4, 12\} = \mathbf{12}$$

The crucial observation here is that the median operation selected different “representative” pixels in each case. When computed separately, the medians came from moderate values (3 and 2). But after addition, the pixel that was smallest in f_1 combined with the largest in f_2 , shifting which pixel becomes the median of the sum.

Step 5: Conclude

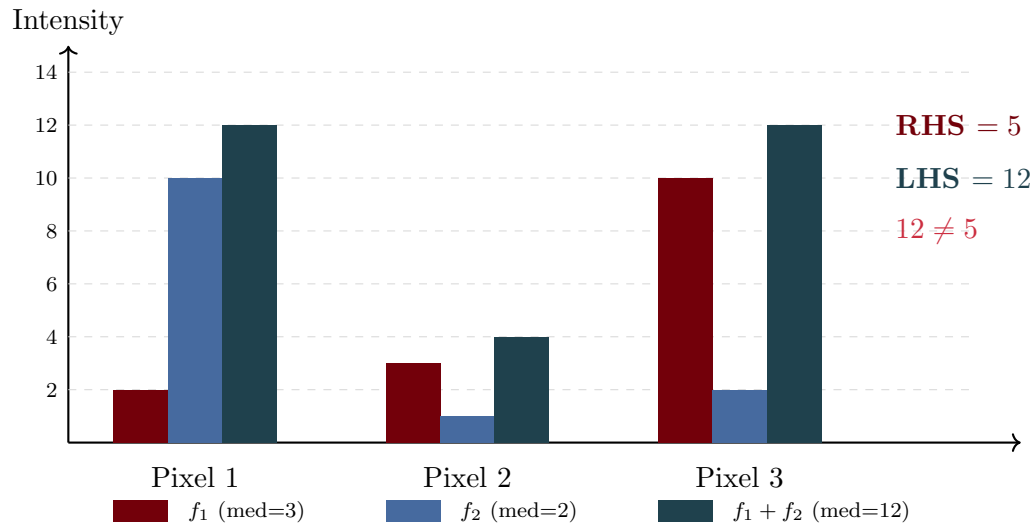
Comparing the two sides of the superposition equation:

$$H[f_1 + f_2] = 12 \quad \text{versus} \quad H[f_1] + H[f_2] = 5$$

Since $12 \neq 5$, the superposition property is violated. This single counterexample is sufficient to prove that the median operator does not satisfy the linearity condition for all inputs. Therefore, the median operator is **nonlinear**.

Intuitively, this occurs because the median is a *rank-based* statistic—it selects the middle value after sorting. When two images are added, the relative ranking of pixels can change dramatically, causing a completely different pixel to be selected as the median. This rank-shuffling behavior is fundamentally incompatible with linearity. ■

Visualization: Counterexample Bar Chart



Results

The median operator is **nonlinear**. The counterexample $f_1 = \{2, 3, 10\}$, $f_2 = \{10, 1, 2\}$ yields:

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2] \quad \blacksquare$$

Problem 2: Image Subtraction for Defect Detection (20 pts)

Image subtraction is used often in industrial applications for detecting missing components in product assembly. The approach is to store a “golden” image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work? You need to list at least two major factors that affect the detection performance.

Demonstration: Golden Image Analysis

To demonstrate these factors concretely, I applied various perturbations to a “golden” reference image of an industrial product (a pharmaceutical tablet) and computed the resulting difference images. Figure 1 shows the golden reference image used for this analysis.



Figure 1: Golden reference image of a pharmaceutical tablet with “FF” marking, used as the baseline for defect detection analysis.

Factors Affecting Detection Performance

1. **Illumination Intensity.** Changes in brightness between the golden and test images cause uniform intensity shifts across the frame. Even ± 40 level variations produce difference signals that overwhelm subtle defects. Fixed, calibrated light sources with stable power are required.
2. **Spatial Translation.** If the product is not positioned identically to the golden capture, even 5-pixel shifts create prominent edge artifacts. Mechanical fixtures and sub-pixel alignment are essential to prevent positional mismatches from triggering false positives.
3. **Rotational Misalignment.** When the product rotates slightly relative to the reference, arc-shaped difference patterns emerge around curved features. A 2° rotation produces detectable artifacts, requiring precise alignment fixtures or rotation correction.
4. **Scale and Zoom Variation.** Differences in camera distance or focal length cause the test image to appear larger or smaller than the reference. A 3% scale change creates ring-like edge differences, requiring fixed working distances and locked zoom.
5. **Sensor Noise.** Random electronic noise introduces speckle patterns throughout the difference image. Gaussian noise with $\sigma = 20$ can mask small defects, making low-noise sensors and temporal averaging important countermeasures.
6. **Focus Blur.** When the test image is out of focus compared to the sharp reference, edges become softer and produce haloed differences. Detection requires auto-focus locks or telecentric lenses for consistent focus.
7. **Non-uniform Illumination.** Vignetting causes corners and edges to appear darker than the center. This spatial gradient creates systematic differences even when the product is identical, requiring flat-field correction or uniform lighting.
8. **Contrast and Gain Variation.** Changes in camera gain or lighting ratios alter the dynamic range of captured images. A 30% contrast shift distorts intensity differences, so fixed settings with auto-gain disabled are necessary.
9. **Color Temperature.** Shifts between warm and cool illumination alter color channel balance. If the golden image used daylight lighting but production uses warmer LEDs, systematic RGB offsets appear, requiring fixed white balance.
10. **Perspective Distortion.** Changes in viewing angle cause trapezoidal warping of the product's appearance. This distortion cannot be corrected by simple translation or rotation, requiring telecentric optics or fixed camera mounts.
11. **Exposure Variation.** Differences in shutter speed or ambient light cause non-linear brightness shifts affecting highlights and shadows. Consistent detection requires fixed exposure and controlled ambient lighting.
12. **Directional Shadows.** Shadows from the product or nearby objects create intensity gradients that vary with lighting position. These overlap with defect locations, making diffuse lighting or light tents essential.

Failure Modes Summary

Figure 2 shows a summary of six key failure modes applied to the golden image. Each column shows: (1) the perturbed test image, (2) the resulting difference image (brighter = larger difference), and (3) the false positives detected at a fixed threshold. Notice how environmental variations that leave the product physically unchanged still trigger significant detection artifacts.

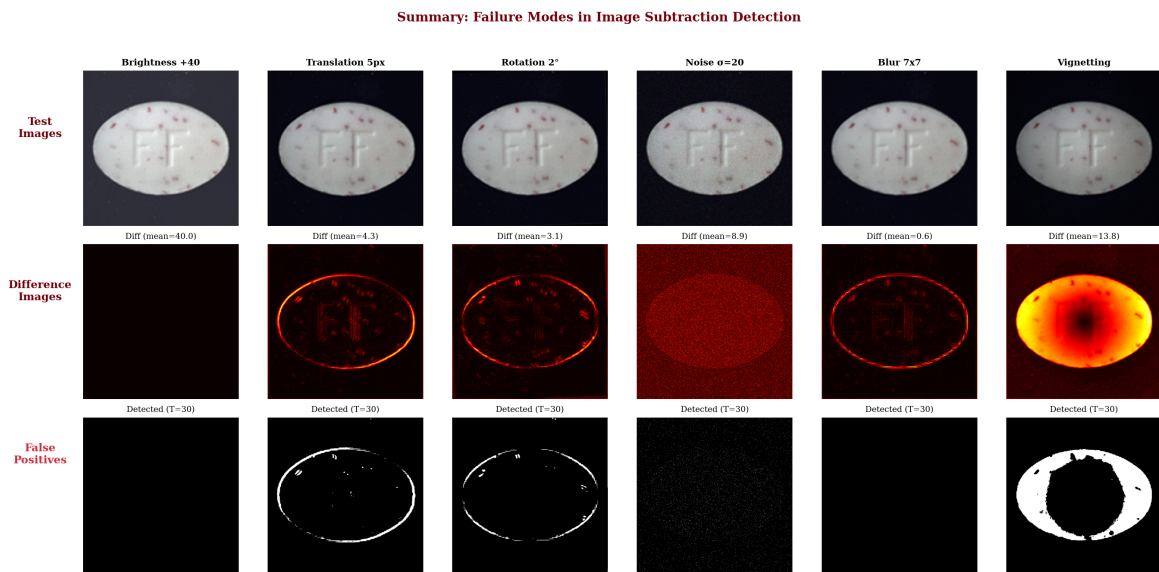


Figure 2: Summary of failure modes in image subtraction detection. Row 1: Test images with various perturbations. Row 2: Difference images (hotter = larger difference). Row 3: Detected regions at threshold $T = 30$. All perturbations produce false positives despite no actual defects.

Quantitative Comparison

Figure 3 quantitatively compares all twelve failure modes by mean absolute difference. The key insight is that **illumination-related factors dominate**: brightness changes and exposure variations cause the largest false positive signals. Geometric factors (translation, rotation, scale) cause smaller mean differences but concentrate at edges where defects are often located. Critically, a true defect (missing component) has a relatively **small** mean difference because it affects only a localized region—demonstrating how failure modes can mask real defects.

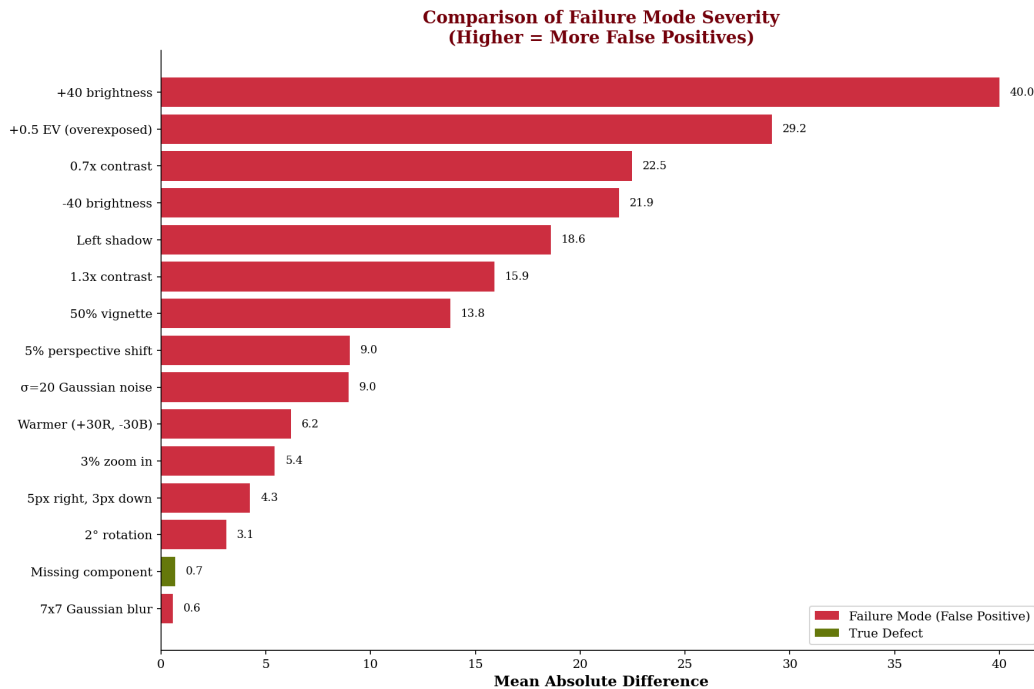


Figure 3: Comparison of failure mode severity measured by mean absolute difference. Higher values indicate more false positives. Note that the true defect (missing component, green) has a *lower* mean difference than most failure modes, illustrating how environmental variations can mask genuine defects.

Results

Major factors affecting image subtraction defect detection:

Primary Factors (Must Control):

1. **Illumination consistency:** Intensity, direction, uniformity, and color temperature must be identical.
2. **Spatial registration:** Position, orientation, and scale must match exactly.

Secondary Factors (Should Control):

3. Sensor noise (requires thresholding)
4. Focus/sharpness consistency
5. Camera gain and exposure settings
6. White balance / color calibration
7. Perspective and optical distortion
8. Shadow elimination

Problem 3: Histogram Equalization Idempotency (20 pts)

Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image) will produce exactly the same result as the first pass.

Step 1: Define the Equalization Transform

Let the original image have intensity levels $r \in [0, L - 1]$ with PDF $p_r(r)$ and CDF $P_r(r)$. The histogram equalization transformation is:

$$s = T(r) = (L - 1) P_r(r) = (L - 1) \int_0^r p_r(w) dw$$

Step 2: Show the Equalized Image Is Uniform

A fundamental result from probability theory: if $s = T(r) = (L - 1) P_r(r)$, then s is **uniformly distributed** on $[0, L - 1]$:

$$p_s(s) = \frac{1}{L - 1}, \quad 0 \leq s \leq L - 1$$

Step 3: Apply Equalization Again

Apply histogram equalization a second time to the equalized image:

$$v = T_2(s) = (L - 1) P_s(s) = (L - 1) \int_0^s p_s(w) dw$$

Since $p_s(s) = \frac{1}{L - 1}$, the CDF is:

$$P_s(s) = \int_0^s \frac{1}{L - 1} dw = \frac{s}{L - 1}$$

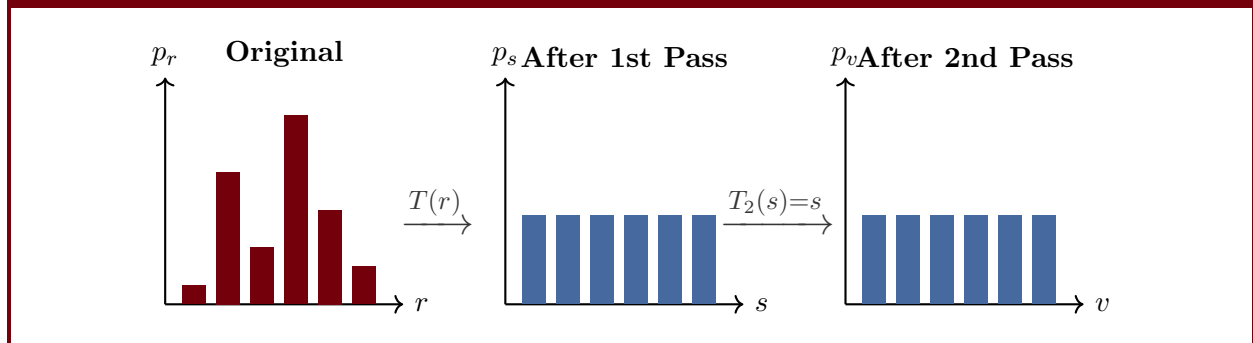
Step 4: Simplify to Identity

Substituting:

$$v = T_2(s) = (L - 1) \cdot \frac{s}{L - 1} = s$$

The second equalization is the **identity transformation**: $v = s$. The output is unchanged. ■

Visualization: Histogram Through Two Equalization Passes



Results

After the first histogram equalization, the output has a uniform distribution. Applying histogram equalization to a uniformly distributed image yields the identity mapping $v = s$. Therefore, the second pass produces exactly the same result as the first. ■

Problem 4: Histogram Matching (30 pts)

Suppose that a 4-bit image ($L = 16$) of size 64×64 has the intensity distribution as Table A. It is desired to transform the original histogram to a specified histogram as shown in Table B. What will be the actual histogram after the histogram matching operation?

r_k	n_k
0	16
1	0
2	512
3	32
4	64
5	128
6	64
7	1024
8	512
9	512
10	16
11	1024
12	0
13	128
14	0
15	64

Table A

z_q	n_q
0	64
1	128
2	512
3	512
4	1024
5	1024
6	512
7	128
8	128
9	64
10	0
11	0
12	0
13	0
14	0
15	0

Table B
Step 1: Setup

Total pixels: $N = 64 \times 64 = 4096$, intensity levels: $L = 16$, so $L - 1 = 15$.
The equalization transform is:

$$s_k = \text{round} \left(15 \cdot \sum_{j=0}^k \frac{n_j}{4096} \right)$$

Step 2: Cumulative Histogram and s_k Values from Table A

r_k	n_k	$\sum_{j=0}^k n_j$	$s_k = \text{round}\left(\frac{15 \cdot \sum_{j=0}^k n_j}{4096}\right)$
0	16	16	$\text{round}(0.059) = 0$
1	0	16	$\text{round}(0.059) = 0$
2	512	528	$\text{round}(1.934) = 2$
3	32	560	$\text{round}(2.051) = 2$
4	64	624	$\text{round}(2.285) = 2$
5	128	752	$\text{round}(2.754) = 3$
6	64	816	$\text{round}(2.988) = 3$
7	1024	1840	$\text{round}(6.738) = 7$
8	512	2352	$\text{round}(8.613) = 9$
9	512	2864	$\text{round}(10.488) = 10$
10	16	2880	$\text{round}(10.547) = 11$
11	1024	3904	$\text{round}(14.297) = 14$
12	0	3904	$\text{round}(14.297) = 14$
13	128	4032	$\text{round}(14.766) = 15$
14	0	4032	$\text{round}(14.766) = 15$
15	64	4096	$\text{round}(15.000) = 15$

Step 3: Cumulative Histogram and $G(z_q)$ Values from Table B

z_q	n_q	$\sum_{j=0}^q n_j$	$G(z_q) = \text{round}\left(\frac{15 \cdot \sum_{j=0}^q n_j}{4096}\right)$
0	64	64	$\text{round}(0.234) = 0$
1	128	192	$\text{round}(0.703) = 1$
2	512	704	$\text{round}(2.578) = 3$
3	512	1216	$\text{round}(4.453) = 4$
4	1024	2240	$\text{round}(8.203) = 8$
5	1024	3264	$\text{round}(11.953) = 12$
6	512	3776	$\text{round}(13.828) = 14$
7	128	3904	$\text{round}(14.297) = 14$
8	128	4032	$\text{round}(14.766) = 15$
9	64	4096	$\text{round}(15.000) = 15$
10	0	4096	15
11	0	4096	15
12	0	4096	15
13	0	4096	15
14	0	4096	15
15	0	4096	15

Step 4: Inverse Mapping $s_k \rightarrow z_q$

For each value of s_k , find the smallest z_q such that $G(z_q) \geq s_k$:

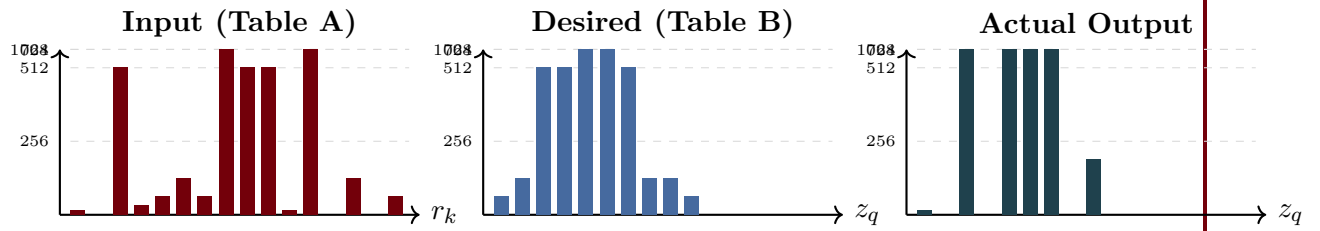
r_k	s_k	Smallest z_q with $G(z_q) \geq s_k$	z_q
0	0	$G(0) = 0 \geq 0$	0
1	0	$G(0) = 0 \geq 0$	0
2	2	$G(2) = 3 \geq 2$	2
3	2	$G(2) = 3 \geq 2$	2
4	2	$G(2) = 3 \geq 2$	2
5	3	$G(2) = 3 \geq 3$	2
6	3	$G(2) = 3 \geq 3$	2
7	7	$G(4) = 8 \geq 7$	4
8	9	$G(5) = 12 \geq 9$	5
9	10	$G(5) = 12 \geq 10$	5
10	11	$G(5) = 12 \geq 11$	5
11	14	$G(6) = 14 \geq 14$	6
12	14	$G(6) = 14 \geq 14$	6
13	15	$G(8) = 15 \geq 15$	8
14	15	$G(8) = 15 \geq 15$	8
15	15	$G(8) = 15 \geq 15$	8

Step 5: Build the Output Histogram

Using the mapping $r_k \rightarrow z_q$ and the pixel counts n_k from Table A:

Output z_q	Mapped from r_k	Pixel count
0	r_0, r_1	$16 + 0 = 16$
1	—	0
2	r_2, r_3, r_4, r_5, r_6	$512 + 32 + 64 + 128 + 64 = 800$
3	—	0
4	r_7	1024
5	r_8, r_9, r_{10}	$512 + 512 + 16 = 1040$
6	r_{11}, r_{12}	$1024 + 0 = 1024$
7	—	0
8	r_{13}, r_{14}, r_{15}	$128 + 0 + 64 = 192$
9–15	—	0 each
Total:		$16 + 800 + 1024 + 1040 + 1024 + 192 = 4096 \checkmark$

Visualization: Input



Results

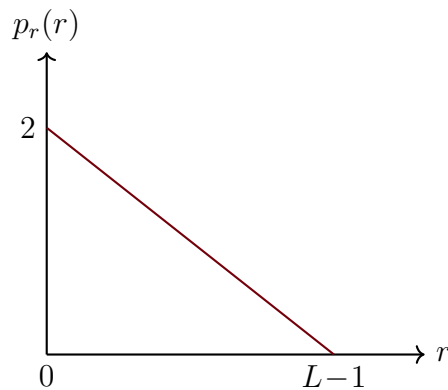
The actual output histogram after histogram matching is:

z_q	n_q^{out}
0	16
1	0
2	800
3	0
4	1024
5	1040
6	1024
7	0
8	192
9	0
10	0
11	0
12	0
13	0
14	0
15	0

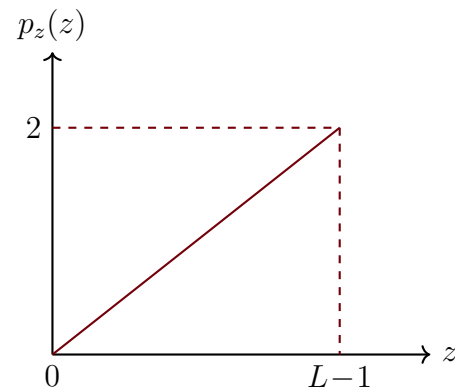
The output histogram approximates the desired distribution but is not identical to it, due to the discrete nature of the intensity levels and the many-to-one mapping inherent in histogram matching.

Problem 5: Continuous Intensity Transformation (20 pts)

An image with intensities in the range $[0, L - 1]$ has the pdf $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this.



(a)



(b)

Step 1: Write $p_r(r)$ with Normalization Verification

Let $W = L - 1$. From Figure (a), $p_r(r)$ is a line from $(0, \frac{2}{W})$ to $(W, 0)$:

$$p_r(r) = \frac{2}{W} \left(1 - \frac{r}{W} \right) = \frac{2(W - r)}{W^2}, \quad 0 \leq r \leq W$$

$$\text{Verification: } \int_0^W \frac{2(W - r)}{W^2} dr = \frac{2}{W^2} \left[Wr - \frac{r^2}{2} \right]_0^W = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 2: Write $p_z(z)$ with Normalization Verification

From Figure (b), $p_z(z)$ is a line from $(0, 0)$ to $(W, \frac{2}{W})$:

$$p_z(z) = \frac{2z}{W^2}, \quad 0 \leq z \leq W$$

$$\text{Verification: } \int_0^W \frac{2z}{W^2} dz = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 3: Compute CDF $T(r)$

$$T(r) = \int_0^r p_r(w) dw = \int_0^r \frac{2(W-w)}{W^2} dw = \frac{2}{W^2} \left[Wr - \frac{r^2}{2} \right] = \frac{2Wr - r^2}{W^2}$$

Step 4: Compute CDF $G(z)$

$$G(z) = \int_0^z p_z(w) dw = \int_0^z \frac{2w}{W^2} dw = \frac{z^2}{W^2}$$

Step 5: Set $G(z)$

$$\begin{aligned} \frac{z^2}{W^2} &= \frac{2Wr - r^2}{W^2} \\ z^2 &= 2Wr - r^2 = r(2W - r) \\ z &= \sqrt{r(2(L-1) - r)} \end{aligned}$$

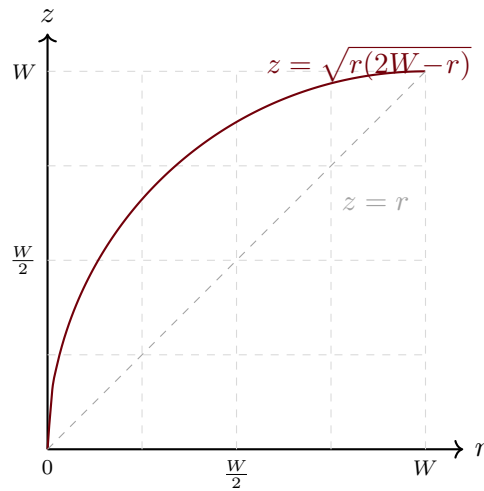
where we take the positive root since $0 \leq r \leq W$ ensures the argument is non-negative.

Step 6: Verify Boundary Conditions

- $f(0) = \sqrt{0 \cdot (2W - 0)} = 0 \checkmark$
- $f(W) = \sqrt{W(2W - W)} = \sqrt{W^2} = W = L - 1 \checkmark$

The transformation maps $[0, L - 1] \rightarrow [0, L - 1]$ as required.

Visualization: Transformation z



The curve is a quarter-circle arc from $(0,0)$ to (W,W) , lying above the identity line, indicating the transformation boosts lower intensities.

Results

The transformation that maps the input PDF $p_r(r)$ to the desired output PDF $p_z(z)$ is:

$$z = f(r) = \sqrt{r(2(L-1) - r)}, \quad 0 \leq r \leq L-1$$

This is a quarter-circle arc from $(0,0)$ to $(L-1, L-1)$. ■